

A Large-Scale Human Toxicogenomics Resource for Drug-Induced Liver Injury Prediction

Corresponding Author: Dr Volker Bergen

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors developed a machine learning model trained by a large-scale and multivariable dataset to accurately predict DILI and the underlying mechanisms. The performance of the model is impressively good, surpassing most of the relevant model. Especially the showcase of DILI prediction for the new drug candidate in clinical trial which failed to be revealed by preclinical tastings showed the promise in translational medicine and drug discovery.

Here are some comments and questions that the authors might want to consider for further improvements:

1. The model was trained by RNA-seq data of sandwich-cultured primary human hepatocytes with a short-term exposure (24 hours). The severe distortions of two-dimensionally cultured hepatocyte gene expression due to de-differentiation has been clearly reported. Moreover, a short time exposure might not be sufficient to mimic many idiosyncratic toxicities that typically occur only after multiple weeks to months. Will such a selection bias of training data cause biased mechanistic prediction of the DILI candidates? This should at least be discussed.
2. In page 3, There seems to be a typo in line 2: "... the transcriptomic first DILI dose...". The word "transcriptomic" seems to be extra here.
3. Why did the authors set the safety margin at 80? Is it an empirical value? Please specify the rationale of setting this value as cut-off since SMs of 30 are chosen in many other studies.
4. In the section of "Known and novel genes and pathways associated with DILI risk", the authors used heatmaps to show AUC of differentially regulated pathways and DILI risk probability. There is a strong positive correlation between differential pathway signature and DILI risk probability. However, the differential pathway signature in the area of high DILI risk probability seems less clear on the heatmap. It would help if the authors could perform correlation analysis to quantify the strength of the relationship.
5. In the section of safety margin prediction for the drugs in clinical trial, which Cmax did the author used, total Cmax or free drug Cmax? Is there a difference which value is used? Please clarify and discuss.
6. Notably, in the map of toxicogenomic prediction tools, the predictive power of the two models trained by 3D-cultured cells showed great performance even with much less input data. I suggest verifying the model with some 3D-cultured cells to show the compatibility of the model in 3D-cultured cells.
7. The code on Github and the method sections were not accessible

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

DILImap – A Comprehensive Large-Scale Human Toxicogenomics Database for Prediction of Drug-Induced Liver Injury

The authors have produced a very comprehensive and interesting paper. However I have the following concerns:

1. The scale of the DILImap is impressive but not particularly novel (bulk seq on primary human hepatocytes).
2. The presence of only one cell type in the assay (PHH) limits the translational impact as DILI is often a result of complex

interplay between many cell types

3. The detail regarding the models used for analysis is limited (presumably for commercial reasons) which limits the value of the paper to the reader.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

Bergen et al. presented DILImap, a large-scale human toxicogenomics resource, along with ToxPredictor, a Random Forest-based framework for predicting drug-induced liver injury (DILI). The authors present a compelling case for their approach, highlighting the limitations of current DILI prediction methods and emphasizing the potential of DILImap and ToxPredictor to address these challenges. The authors demonstrated the model's ability to achieve high sensitivity and specificity in validation using an independent dataset, outperforming state-of-the-art pre-clinical models. Notably, their approach also flagged previously overlooked failed trials due to DILI, such as Evobrutinib, TAK-875, and BMS-986142. While the study presents an important resource and framework for DILI prediction, several concerns warrant clarification.

1. The numbers of drugs are not consistent and confusing across the manuscript and supplementary data files. While DILImap was claimed to include 300 drugs, the Results section mentions 177 DILI+ and 93 DILI- in the training set and 32 DILI+ and 14 DILI- in the validation data, which are 316 drugs. Supplementary Table shows a total of 299 drugs "DILIMAP_RESULTS".

2. In the modeling process, it is unclear whether the cross-validation mode was done for using all drugs in DILImap or only the training data.

3. No details on how the training and validation datasets were constructed/split are given. What approach was taken to minimize potential biases in the validation set? This information would be particularly important with the claimed 100% specificity.

4. Despite extensive comparisons with various assays, the study does not benchmark ToxPredictor against other computational models, such as DILIPredictor, DILIGeNN, etc, for the same drugs.

5. No rationale or supporting data were presented when choosing random forest as the algorithm.

6. Why 30-fold cross-validation was chosen? This fold seems very high considering the small dataset of 300 drugs.

7. An explanation of each column in supplementary tables would be useful.

8. Codes and data files were available through figshare and code ocean but not completely with many missing data files (e.g. DILImap_deseq2.h5ad). Please ensure all data files are provided.

(Remarks on code availability)

The authors provided online documentation for installing and using the package; however, the installation instructions did not work with the current version available in their private GitHub repository. While most code and data were accessible, installing the package was challenging. Additionally, some data files were missing, preventing the successful execution of certain scripts.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have successfully addressed the concerns and questions I raised.

It is appreciated that the entire codebase, data, and user guide are now accessible via GitHub, making it easy for users to install the PyPI package. Furthermore, the missing method section is now available in the supplementary file.

I also appreciate the efforts of the authors to clarify the application of free Cmax or total Cmax. The author demonstrated that neither free Cmax nor total Cmax significantly impacted their model's ability to predict DILI. This is very interesting!

It is also good to see that the authors discussed the combination of emerging 3D liver models and DILImap in the updated manuscript. This is very important to improve the future application of New Approach Methodologies for drug development.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

I am satisfied with the authors comments

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors have addressed most of my previous comments sufficiently, and I appreciate the clarifications provided in the revision. I have only a few minor follow-up comments.

1. The final rationale for prioritizing Random Forest over other models with similar AUCs could still be made clearer. It would help to briefly explain how the evaluation metrics such as AUC, inter-fold correlation, dose response, precision, and recall were weighted or prioritized in arriving at the final choice.
2. In Figure S4C, only performance metrics are shown without the corresponding hyperparameters. Since it is not feasible to include various hyperparameters in this table, I suggest providing another supplementary file with hyperparameter details for these model configurations.
3. Regarding the CV clarification, I understand from the revised text that 5-fold stratified compound-level cross-validation (30-fold in the original submission) was used for model training and evaluation. The compound-level approach, described as “all samples from a single compound were held out together,” is clear and appropriate for preventing data leakage. The exact number of folds better be stated explicitly in the Methods section as well.
4. The current description in the final RF ensemble model, “The final model is an ensemble of 30 Random Forests trained across cross-validation folds to enhance generalization and prediction stability,” is confusing. It would be clearer to use the original wording “30 models” or “30 ensemble members,” rather than the current “30 folds.” (here and in the Methods as well).

(Remarks on code availability)

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Point-by-point Response to the Reviewers

We thank the reviewers for their thoughtful feedback. We have carefully addressed all comments and substantially revised the manuscript. Key updates include:

- Public release of the full codebase via GitHub and PyPI for easy installation
- Complete availability of validation and processed training data for full reproducibility
- Additional experiments, including evaluation of free vs. total Cmax
- Benchmarking against state-of-the-art in silico models
- Detailed comparison with 3D liver models
- All manuscript changes are highlighted in yellow for ease of review.

To simplify access and ensure full reproducibility:

- **Results, notebooks, and setup instructions** are available at dilimap.org/review-dUFZuIWv8k7bERJ3FQs4, which will be changed to dilimap.org upon publication. A [ZIP archive](#) of all reproducibility notebooks is also available for download.
- **Code** is hosted publicly at github.com/Cellarity/DILImap.
- **Data and trained models** are accessible via an S3 bucket for seamless use in Jupyter notebooks or scripts. Reviewers can use the credentials below for access during peer review. Upon publication, these credentials will no longer be needed, as the data will be freely available and also deposited in GEO. A detailed dataset description is available [here](#). Only raw training data (raw counts) are proprietary and available upon request, subject to data-sharing agreements, while processed training data (pathway-level signatures used as model input) are provided to ensure full reproducibility of model training and all figures.

Credentials to access data:

```
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AWS_SECRET_ACCESS_KEY = CCUSUm/oT5B0A/T7+Lum74Zkq6a85o64Btj+1WRx
```

We believe these revisions significantly strengthen the manuscript and fully address the key concerns regarding novelty, reproducibility, and benchmarking.

Please find our detailed point-by-point responses in the following.

Reviewer #1 (remarks to the author)

The authors developed a machine learning model trained by a large-scale and multivariable dataset to accurately predict DILI and the underlying mechanisms. The performance of the model is impressively good, surpassing most of the relevant model. Especially the showcase of DILI prediction for the new drug candidate in clinical trial which failed to be revealed by preclinical tastings showed the promise in translational medicine and drug discovery.

Here are some comments and questions that the authors might want to consider for further improvements:

1. The model was trained by RNA-seq data of sandwich-cultured primary human hepatocytes with a short-term exposure (24 hours). The severe distortions of two-dimensionally cultured hepatocyte gene expression due to de-differentiation has been clearly reported. Moreover, a short time exposure might not be sufficient to mimic many idiosyncratic toxicities that typically occur only after multiple weeks to months. Will such a selection bias of training data cause biased mechanistic prediction of the DILI candidates? This should at least be discussed.

Response:

Thank you for raising this important point. We agree that 2D culture systems, including sandwich-cultured primary human hepatocytes (SC-PHHs), have limitations, particularly in modeling long-term or immune-mediated toxicities. We now acknowledge this in the Discussion section.

Regarding hepatocyte de-differentiation, our choice of a 24-hour exposure time represents a compromise between biological relevance and practical constraints related to cell viability and transcriptomic integrity. To address concerns about cellular identity, we have added Suppl. Figure S2, which shows consistent expression of key hepatocyte markers and low expression of marker genes for non-parenchymal liver cells (Kupffer cells, LSECs, hepatic stellate cells, cholangiocytes) or de-differentiation markers, supporting preservation of hepatocyte phenotype with minimal contamination or transdifferentiation.

Addition to Manuscript:

Results section

“All compounds were screened in sandwich-cultured primary human hepatocytes (PHHs), the gold standard and most physiologically relevant in vitro model for liver toxicity, which preserve key hepatic functions such as metabolic activity and bile canaliculi formation^{38,39} [...] We selected a 24-hour post-exposure time point, based on the trade-off between signal strength and cellular viability: earlier time points (e.g., 2h or 8h) yield weaker transcriptional responses⁴⁰, while longer incubations risk hepatocyte de-differentiation and RNA degradation⁴¹. Marker analysis confirmed retention of hepatocyte identity at 24 hours. We further ensured data quality by

including only wells with sufficient total RNA counts and low mitochondrial RNA content, indicating viable, transcriptionally active cells (**Suppl. Fig. S2**)."

Discussion section

"[...] Fourth, the exposure timepoint of 24 hours limit the detection of delayed or immune-mediated toxicities."

"Looking ahead, toxicogenomics holds tremendous promise for advancing our understanding and prediction of DILI. Integrating RNA-seq with advanced 3D culture systems—such as spheroids and liver-on-chip platforms—may enable longer drug exposures and the inclusion of non-parenchymal cells like Kupffer cells and hepatic stellate cells. These co-culture models are essential for capturing immune activation, inflammation, and fibrosis—hallmarks of idiosyncratic and chronic DILI that hepatocyte monocultures cannot recapitulate. As long 3D models rely on high-content imaging or cytotoxicity endpoints that capture specific phenotypic responses they remain limited in mechanistic scope and scalability. Their performance is often demonstrated on small, curated compound sets with known mechanisms, which may not translate to broader chemical space. In contrast, transcriptomics provides a scalable, unbiased readout capable of detecting diverse DILI mechanisms, including those not captured by existing assays. As RNA-seq becomes more feasible in physiologically relevant 3D systems, we anticipate a powerful synergy—combining the mechanistic breadth of transcriptomics with the physiological relevance of 3D models. Our work lays the foundation for such integration, demonstrating that transcriptomics alone can robustly capture DILI risk across a wide range of mechanisms."

Suppl. Figure S2 – panel C

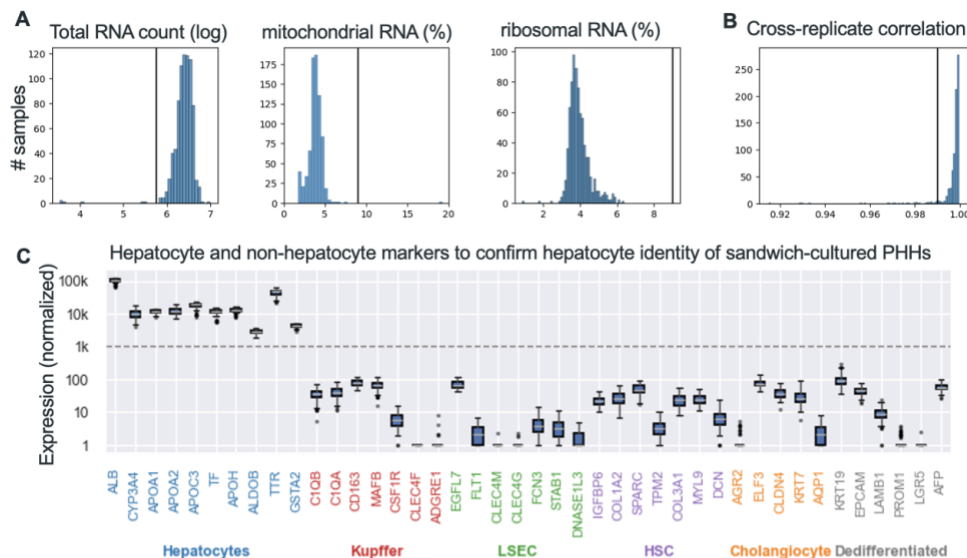


Figure S2. Quality control and confirmation of hepatocyte identity in sandwich-cultured primary human hepatocytes (PHHs). **A.** Distribution of RNA quality metrics across samples: total RNA count (log scale), mitochondrial RNA fraction, and ribosomal RNA fraction. QC thresholds used for sample inclusion were set at ~700k total counts (~2SD above mean), <9% mitochondrial RNA and <9% ribosomal RNA < 9% (black vertical lines). **B.** Cross-replicate correlation across all RNA-seq profiles, with only highly reproducible samples (Pearson $r \geq 0.99$) retained. **C.** Expression of canonical marker genes confirms hepatocyte identity. Hepatocyte-specific markers show high expression (>2,000 counts), while markers for non-hepatocyte liver cell types (Kupffer cells, liver sinusoidal endothelial cells, hepatic stellate cells, cholangiocytes) and dedifferentiation markers are consistently low (<100 counts), supporting purity and preserved functional hepatocyte function at 24 hours post-treatment.

2. In page 3, There seems to be a typo in line 2: "... the transcriptomic first DILI dose...". The word "transcriptomic" seems to be extra here.

Response: Thank you, we have dropped 'transcriptomic'.

3. Why did the authors set the safety margin at 80? Is it an empirical value? Please specify the rationale of setting this value as cut-off since SMs of 30 are chosen in many other studies.

Response:

Thank you for this important question. The MOS threshold of 80 was selected based on empirical evaluation on held-out training data, optimized to balance sensitivity and specificity. While MOS values around 30 are common in cytotoxicity or mechanistic assays, our model is based on transcriptomic responses, which are more sensitive and often respond at lower concentrations. A higher threshold is therefore appropriate to avoid overcalling risk based on early, potentially adaptive gene expression changes. Importantly, a threshold of 80 remains well within the commonly cited literature range of 10 to 100 (now clarified in both *Results* and *Methods* sections; see additions below).

Addition to Manuscript:

Results section

"A safety margin threshold of 80 provides an actionable classification into high- and low-risk compounds. The probability threshold of 0.7 and MOS cutoff of 80 were both optimized on held-out training data to reach performance plateaus while minimizing false positives. Our selected MOS threshold of 80, while on the higher end of literature-reported ranges (10–100)^{17,23,28,33,35}, reflects the greater sensitivity of transcriptomic assays compared to cytotoxicity or mechanistic readouts. Since transcriptional changes often occur at lower doses—before overt toxicity—a higher cutoff is needed to avoid false positives and maintain specificity in a transcriptome-based model (**Suppl. Fig. S5**).

Methods section

"For validation and benchmarking, to classify a compound as DILI, we selected a Margin of Safety (MOS) threshold of 80, based on empirical evaluation across benchmark datasets. This value reflects an optimized trade-off between sensitivity and specificity—minimizing false positives while reaching a performance plateau that captures the majority of true DILI liabilities. Importantly, this threshold is consistent with values commonly used in the literature, where MOS cutoffs typically range from 10 to 100 depending on context, placing our choice within accepted toxicological standards."

Suppl. Figure S5 – panel B

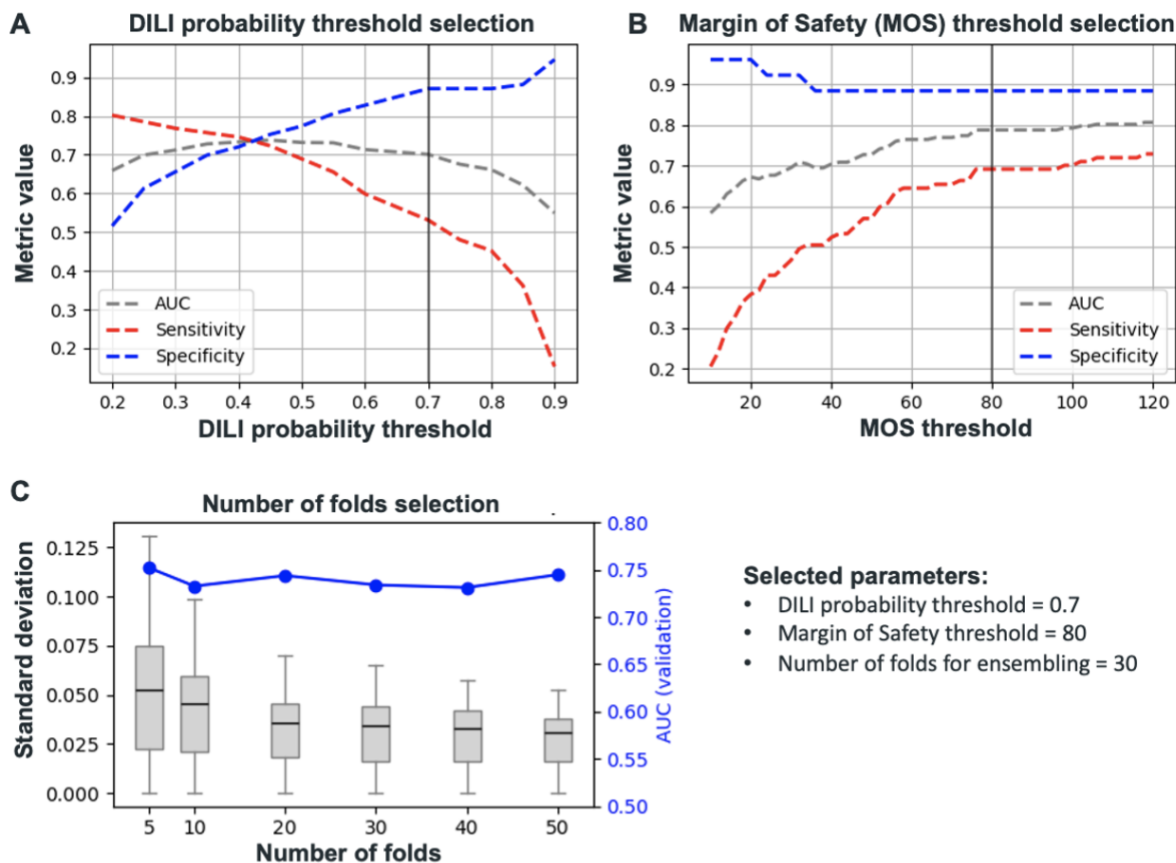


Figure S5. Selection of model threshold parameters based on performance tradeoffs. **A.** DILI probability threshold selection based on sensitivity, specificity, and AUC. A threshold of 0.7 was chosen as it achieves near-plateaued high specificity at acceptable sensitivity. Metric values are computed per-compound/dose basis. Note, that the evaluation metrics here are computed on a per-compound-and-dose basis without inclusion of Cmax. **B.** Margin of Safety (MOS) threshold of 80 was selected for its high plateaued sensitivity and favorable AUC, while preserving high specificity. **C.** Selection of the number of cross-validation folds for model ensembling. Increasing the number of folds reduces the standard deviation across runs, with 30 folds offering a stable tradeoff between ensemble consistency and AUC (validation).

4. In the section of “Known and novel genes and pathways associated with DILI risk”, the authors used heatmaps to show AUC of differentially regulated pathways and DILI risk probability. There is a strong positive correlation between differential pathway signature and DILI risk probability. However, the differential pathway signature in the area of high DILI risk probability seems less clear on the heatmap. It would help if the authors could perform correlation analysis to quantify the strength of the relationship.

Response:

We appreciate this great suggestion. To quantify the relationship between differential pathway signatures and predicted DILI risk, we have added Spearman correlation in Figure 4A. This panel displays the correlation coefficients between pathway activation scores and predicted DILI probabilities across compounds. The analysis shows that pathways related to transcriptional reprogramming and

metabolic stress—such as nuclear receptor signaling, one-carbon metabolism, and bile acid regulation—are among the most strongly correlated with DILI risk. This addition reinforces the mechanistic interpretability of the model's predictions and clarifies the gradient observed in the heatmap

Addition to Manuscript:

Results section

“Additionally, pathway activations highly correlated with predicted DILI risk include *nuclear receptor signaling* (e.g., PXR/CAR/FXR), *one-carbon metabolism*, and *bile acid regulation*—highlighting transcriptional reprogramming and metabolic stress as key contributors to hepatotoxicity^{53–55} (Figure 4A). When compounds are ranked by predicted DILI probabilities, a clear gradient of pathway activation emerges, revealing distinct enrichment patterns for these biological processes. This correlation reinforces the direct mechanistic relevance and interpretability of the model's predictions and highlights these pathways as potential drivers of DILI (Figure 4B).

Figure 4A

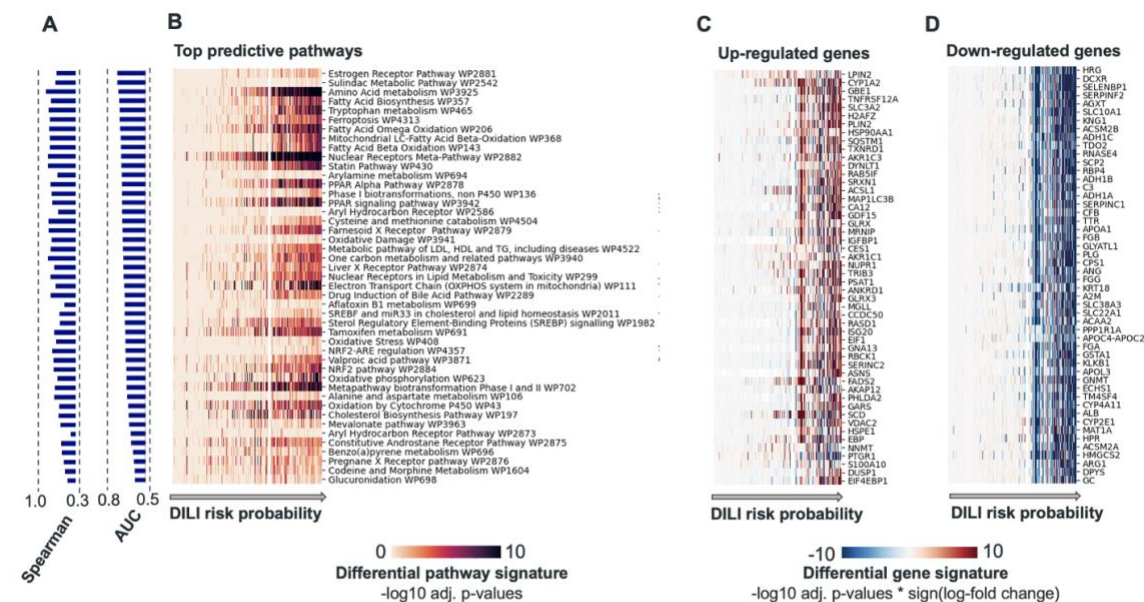


Figure 4. Key pathway activations and genes frequently implicated in DILI. **A.** Pathways are ranked by their ability to distinguish DILI vs. non-DILI compounds, measured by AUC. To quantify the strength of association between pathway activation scores and predicted DILI risk, we show the Spearman correlation between pathway activation scores and DILI probability across compounds. This highlights transcriptional and metabolic stress pathways—such as nuclear receptor signaling, one-carbon metabolism, and bile acid regulation—as key contributors to DILI risk. **B.** Compounds are ordered by their DILI risk probability, showing a correlation between DILI risk and the activation of pathways associated with liver injury. The top affected pathways include those critical to drug metabolism, oxidative stress and lipid homeostasis. Key pathways include Cytochrome P450 Oxidation, NRF2-ARE regulation for oxidative stress response, and Fatty Acid Beta Oxidation, highlighting the interplay between detoxification, mitochondrial function, and energy metabolism.

5. In the section of safety margin prediction for the drugs in clinical trial, which Cmax did the author used, total Cmax or free drug Cmax? Is there a difference which value is used? Please clarify and discuss.

Response:

Thank you for raising this important point. In our analysis, we used total Cmax rather than free Cmax, as total Cmax values were more broadly available. To address the potential impact of this choice, we compared both measures and found that they yield highly correlated safety margin (MOS) estimates and nearly identical predictive performance (see Suppl. Figure S5).

Addition to Manuscript:

Results section

“Among available exposure measures, we used total Cmax instead of free Cmax due to its broader availability across compounds. Both measures yielded nearly identical predictive performance and highly correlated MOS values (**Suppl. Fig. S6**).”

Suppl. Figure S6

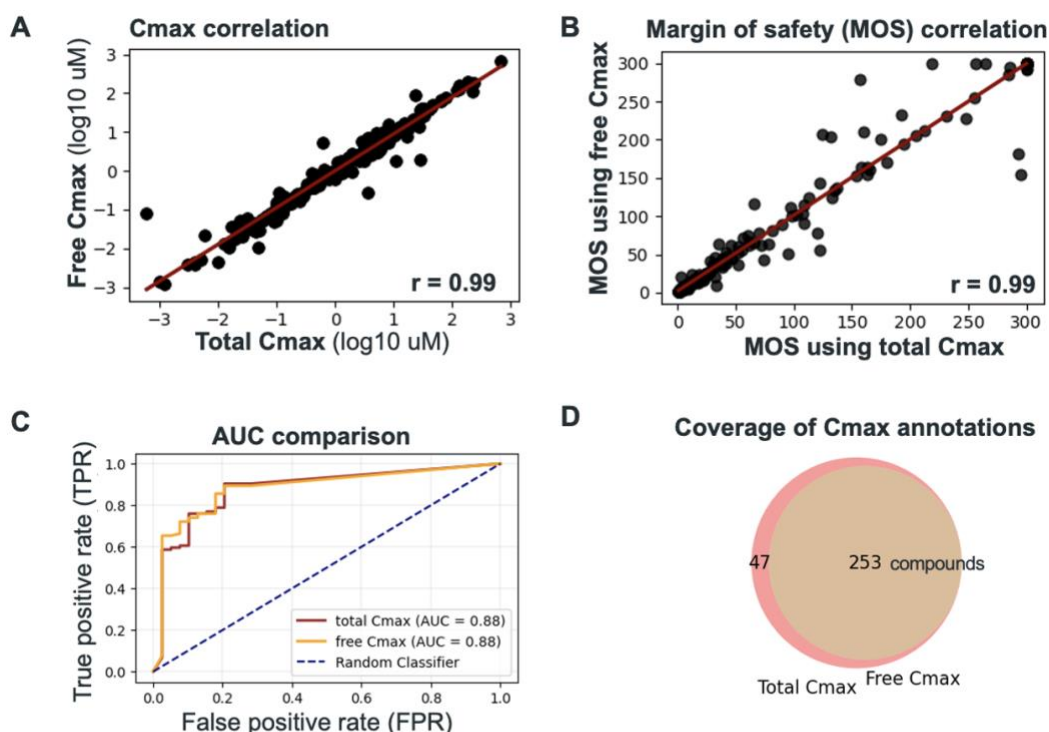


Figure S6. Free vs. Total Cmax: Predictive Value and Availability. **A.** Free and total Cmax values are highly correlated (Pearson $r = 0.99$; log-transformed). **B.** C Margin of Safety (MOS) values derived from free and total Cmax also show near-perfect agreement ($r = 0.99$). **C.** ROC analysis demonstrates similar classification performance using either free Cmax or total Cmax (AUC = 0.88) for compounds with both values available. **D.** Annotation coverage is more limited for free Cmax, with 253 of 300 compounds annotated. Due to its broader availability, total Cmax is used for safety margin estimation and downstream analyses.

6. Notably, in the map of toxicogenomic prediction tools, the predictive power of the two models trained by 3D-cultured cells showed great performance even with much less input data. I suggest verifying the model with some 3D-cultured cells to show the compatibility of the model in 3D-cultured cells.

Response:

We thank the reviewer for this important comment. We address it in two parts:

1. **On the performance of 3D-cultured models:**

We agree that 3D hepatocyte models using cytotoxicity or high-content imaging (HCI) endpoints can show strong performance, even with limited data—as also shown in our benchmarks. However, these readouts are direct measurements that require no model training but are inherently restricted to the specific biological processes they capture. As a result, performance tends to be high in well-characterized carefully selected compound sets but are unlikely to hold true on a dataset at scale like DILImap with broader and mechanistically diverse chemistries. While effective in that narrow context, they may not generalize to mechanistically diverse or novel compounds whose mechanism is not captured by the endpoint. In contrast, transcriptomics enables an unbiased, systems-level view of cellular responses and supports generalizable models that detect hepatotoxicity even when mechanisms are unknown or not covered by predefined conventional endpoints.

2. **On model compatibility with 3D transcriptomic data:**

We agree that validating our model on transcriptomic data from 3D-cultured systems would be valuable. While such datasets are not currently publicly available, our model was specifically designed to generalize across conditions. We are optimistic about its applicability to 3D contexts and have added a note in the *Discussion* highlighting this as an important future direction.

Addition to Manuscript:

Results section

“We benchmark our model along two key axes: *predictive performance* and *scalability*. Predictive performance, measured by balanced accuracy, reflects the model’s ability to distinguish DILI-positive from DILI-negative compounds. Scalability captures both technical throughput and biological breadth—the capacity to generalize across diverse chemistries and mechanisms, including previously uncharacterized ones (**Figure 6A**).

[...]

3D liver systems offer important physiological context. High-content imaging in 3D models, such as those by Walker et al.’s³³ and Ewart et al.’s³⁴, achieves similar performance on small, curated panels (Walker: 23/27 vs. 23/27; Ewart: 11/14 vs. 13/14). However, their limited scalability constrains their utility to a broader range of DILI mechanisms. They may perform well on narrow, curated panels, but struggle with unknown mechanisms or mechanisms not captured by the low-dimensional endpoint, as shown in the following comparison. To explore the unique capabilities of 2D transcriptomics vs 3D cytotoxicity assays, we conducted direct compound-level comparisons with larger 3D screening studies: Vorrink et al²⁶ and Fäs et al³⁵. In Vorrink et al., 3D cytotoxicity uniquely detected 3 compounds (Fialuridine,

Methotrexate, Trazodone), all linked to cytotoxic effects that result in acute cell death. Conversely, our model uniquely flagged 10 compounds—including fluconazole, phenytoin, and zileuton—associated with immune activation, metabolic stress, or enzyme modulation, which are not readily captured by viability endpoints. A similar pattern emerged in the Fäs et al. study: 3D cytotoxicity exclusively identified 4 compounds (e.g., Haloperidol, Fialuridine) whose toxicities depend on structural or metabolic context. Our model uniquely identified 5 compounds (e.g., Cimetidine, Fluconazole, Ximelagatran) marked by subtle transcriptional responses rather than overt cell death. These comparisons highlight a key limitation of endpoint-defined models: while effective in narrow contexts, they struggle with broader chemical and mechanistic diversity. Our transcriptomic approach, by contrast, offers systems-level resolution that generalizes across DILI pathways—not only detecting known cytotoxic responses but also uncovering less immediate, non-lethal mechanisms often missed by traditional assays. While 3D systems offer physiological depth, their reliance on fixed endpoints and lower throughput constrains their utility for a broader range of DILI mechanisms (**Suppl. File S4**).”

Discussion section

“Looking ahead, toxicogenomics holds tremendous promise for advancing our understanding and prediction of DILI. Integrating RNA-seq with advanced 3D culture systems—such as spheroids and liver-on-chip platforms—may enable longer drug exposures and the inclusion of non-parenchymal cells like Kupffer cells and hepatic stellate cells. These co-culture models are essential for capturing immune activation, inflammation, and fibrosis—hallmarks of idiosyncratic and chronic DILI that hepatocyte monocultures cannot recapitulate. As long 3D models rely on high-content imaging or cytotoxicity endpoints that capture specific phenotypic responses they remain limited in mechanistic scope and scalability. Their performance is often demonstrated on small, curated compound sets with known mechanisms, which may not translate to broader chemical space. In contrast, transcriptomics provides a scalable, unbiased readout capable of detecting diverse DILI mechanisms, including those not captured by existing assays. As RNA-seq becomes more feasible in physiologically relevant 3D systems, we anticipate a powerful synergy—combining the mechanistic breadth of transcriptomics with the physiological relevance of 3D models. Our work lays the foundation for such integration, demonstrating that transcriptomics alone can robustly capture DILI risk across a wide range of mechanisms.”

7. The code on Github and the method sections were not accessible

Response:

We thank the reviewer for pointing this out. As part of our original submission, we used CodeOcean to ensure reproducibility during peer review, with the intention of releasing the full GitHub repository upon acceptance of the manuscript. However, in response to this request—and to facilitate immediate and unrestricted access—we have now made the GitHub repository publicly available. The code is installable via PyPI and includes detailed instructions and reproducibility notebooks. Relevant links and installation details are now included in the *Code and Data Availability* section.

Additions to manuscript:

Data and code availability section

Results: All results, reproducibility notebooks, and setup instructions are available at dilimap.org/review-dUFZuIWv8k7bERJ3FQs4. This link will be updated to dilimap.org upon publication. All notebooks can also be downloaded as a ZIP archive [here](#).

Code: The full Python implementation is publicly available at github.com/Cellarity/DILImap

Data: All datasets and trained models are accessible via an S3 bucket, enabling seamless integration with Jupyter notebooks or scripts. Reviewers can use the credentials below for access; the data will be made openly available and deposited in GEO under a Creative Commons license upon publication. A detailed dataset description is available [here](#). Please note, that raw training data (raw counts) are proprietary and available upon request, subject to data-sharing agreements, while processed training data (pathway-level signatures used as model input) are provided to ensure full reproducibility of model training and all figures.

Credentials to access data:

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```


Reviewer #2 (remarks to the author)

The authors have produced a very comprehensive and interesting paper. However I have the following concerns:

1. The scale of the DILImap is impressive but not particularly novel (bulk seq on primary human hepatocytes).

Response:

We appreciate the reviewer's perspective. While we agree that bulk RNA-seq in primary human hepatocytes (PHHs) is an established assay format, the novelty and impact of DILImap lie in how it transforms this method into a predictive, generalizable, and mechanistically rich framework for DILI assessment—addressing key limitations of current preclinical models.

First, the scale and diversity of DILImap are unmatched: over 300 clinically annotated drugs profiled across multiple concentrations, providing a dose-resolved, high-fidelity view of hepatocyte transcriptional responses. To our knowledge, no existing dataset combines this depth of exposure profiling with the breadth of compound diversity in PHHs.

Second, DILImap is purpose-built for predictive modeling. From automated solubility testing and IC_{10} -guided dosing to a unified experimental pipeline with matched DMSO controls, the design ensures biological interpretability and modeling robustness. Unlike prior transcriptomic studies that focus on clustering or exploratory analysis, we develop a cross-validated, compound-level prediction model that not only generalizes to unseen compounds but also outperforms both traditional in vitro assays and modern in silico models (e.g., DILIPredictor, DILIGeNN, TxGemma). Importantly, it flags high-risk clinical candidates like *Evobrutinib*, *TAK-875*, and *BMS-986142* that passed preclinical safety assessments yet failed due to liver toxicity in Phase III trials.

Third, we demonstrate the unique mechanistic breadth of transcriptomics, particularly in contrast to 3D liver models. While 3D systems offer physiological relevance, they are often constrained by low-dimensional readouts such as viability or ATP levels. By capturing full-transcriptome responses, our model detects upstream DILI-related pathways—including oxidative stress, mitochondrial dysfunction, and immune-mediated injury—*before* overt cytotoxicity is observed. Direct compound-level comparisons with recent 3D studies (e.g., *Fäs et al.*) reveal that our model identifies non-cytotoxic liabilities systematically missed by 3D viability-based assays.

Finally, we believe our work offers an unprecedented level of benchmarking rigor. We systematically compare ToxPredictor against over 20 preclinical assays across both in vitro and in silico domains, using shared test sets. To the best of our knowledge, this is the most comprehensive benchmarking of DILI prediction models to date, and highlights the distinct strengths of transcriptomics in terms of both predictive performance and scalability.

Taken together, the combination of biological fidelity, experimental rigor, predictive accuracy, and mechanistic insight defines DILImap as a novel and impactful platform for preclinical DILI assessment. We have revised the *Abstract*, *Introduction*, and *Results* to clarify these contributions, and we have expanded the benchmarking and discussion of synergies with emerging 3D systems in the *Discussion* section.

To clarify these points, we have added the following to the manuscript:

Abstract

“[...] Unlike single-endpoint readouts—even from 3D models—transcriptomics offers a multi-dimensional system-level view of hepatocyte responses, capable of detecting diverse DILI mechanisms not captured by conventional assays.”

Introduction section

“3D liver models aim to better mimic human tissue characteristics and *in vivo* responses¹². However, despite improved physiological relevance, these models remain constrained by low-dimensional readouts—typically a limited panel of markers such as ATP levels, LDH release, or imaging-based features—which fail to capture the full spectrum of molecular responses and often miss the mechanistic cause. This gap continues to result in late-stage drug withdrawals and clinical failures, highlighting the urgent need for more comprehensive and predictive DILI models⁹.

[...]

Beyond generalization to unseen compounds, the model has a distinct edge in its mechanistic breadth. The model leverages the full transcriptomic landscape to detect DILI-related mechanisms such as mitochondrial dysfunction, oxidative stress, immune activation, and metabolic perturbation—often well before cell death. These advantages are particularly evident when compared to high-content 3D liver assays^{26,33–35}, which, while physiologically relevant, are typically constrained to low-dimensional viability or imaging endpoints. In head-to-head comparisons, our model uniquely identifies non-cytotoxic risks that 3D assays miss. This systems-level resolution enables more comprehensive and unbiased detection of toxic liabilities across diverse compound classes, demonstrating its utility for prioritizing safer drug candidates.”

Results section

“We benchmark our model along two key axes: *predictive performance* and *scalability*. Predictive performance, measured by balanced accuracy, reflects the model’s ability to distinguish DILI-positive from DILI-negative compounds. Scalability captures both technical throughput and biological breadth—the capacity to generalize across diverse chemistries and mechanisms, including previously uncharacterized ones (**Figure 6A**).

[...]

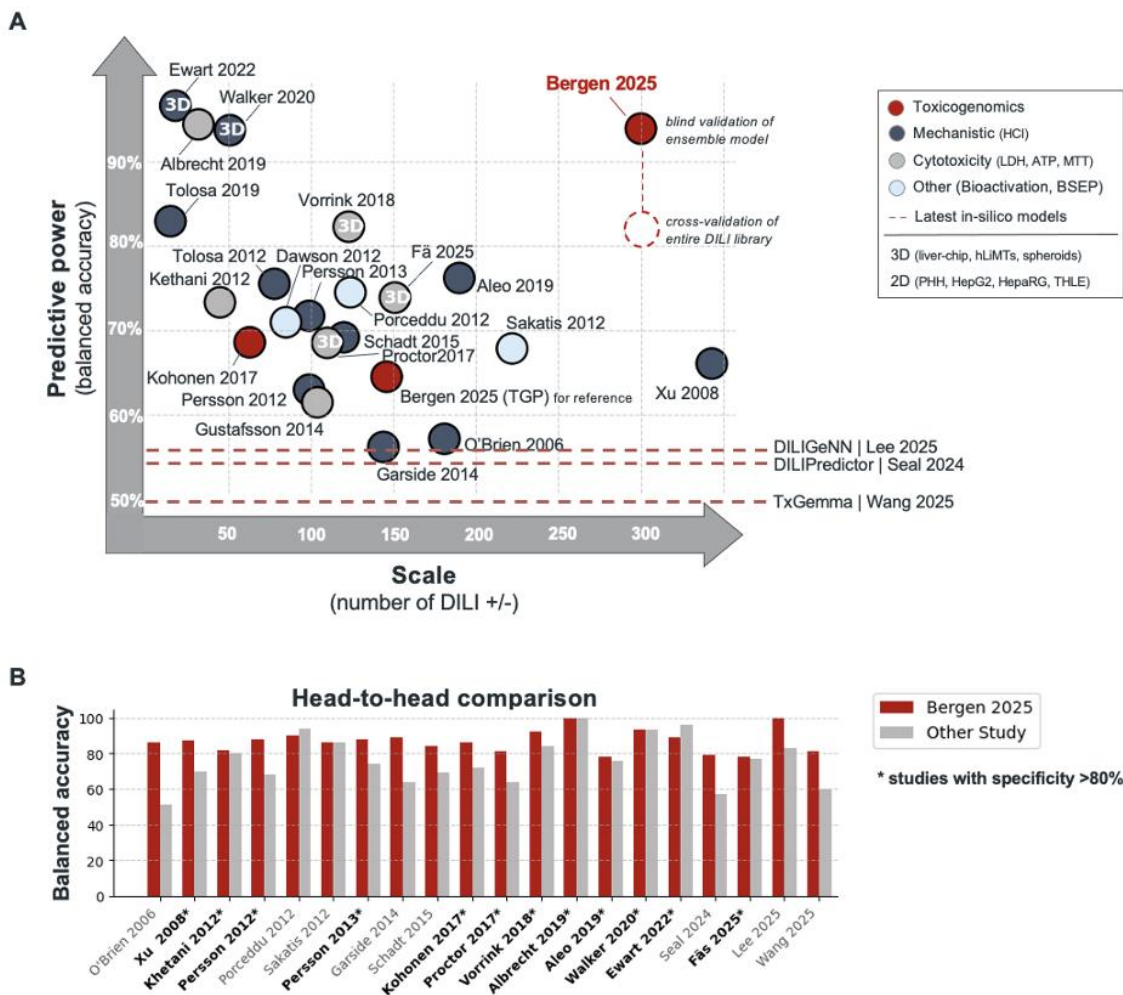
Structure-based in silico models such as TxGemma³¹, DILIGeNN³² and DILIPredictor³³ consistently underperform compared to in vitro–based approaches. To assess real-world generalizability, we evaluated them on 314 independent compounds (45 DILI+, 269 DILI–) primarily annotated via LiverTox (scores A/B as DILI+, E as DILI–); TxGemma was also tested on an expanded set (143 DILI+, 536 DILI–). All showed limited specificity: DILIGeNN (84% sensitivity, 28% specificity), DILIPredictor (80%, 29%), and TxGemma-27B (57%, 37%). These findings are slightly below the balanced accuracy of 0.59 reported by Seal et al. (2024)³³ for DILIPredictor. On a shared benchmark set, TxGemma reached 63% sensitivity (39/62) and 57% specificity (20/35), while our model achieved 76% sensitivity (47/62) and 86% specificity (30/35). DILIGeNN showed perfect sensitivity (5/5) at moderate specificity (2/3), while our model reached 100% on both (5/5 and 3/3). DILIPredictor reached complete sensitivity (23/23) but at the expense of poor specificity (1/7), while ToxPredictor maintained high sensitivity (20/23) at markedly higher specificity (5/7). Low specificity is a key limitation of structure-based models, which lack biological context and tend to over-call toxicity. This results in false positives for commonly prescribed drugs with no risk of hepatotoxicity, such as biotin (DILIPredictor), vitamin D (DILIGeNN), and pemetrexed (all three models). Moreover, they provide only binary outputs, without dose or mechanistic insight. In contrast, transcriptomics enables dose-resolved predictions, mechanistic interpretability, and safety margin estimation—critical for evaluating toxic liabilities and guiding follow-up experiments (**Suppl. Figure S10; Suppl. File S4**).

[...]

3D liver systems offer important physiological context. High-content imaging in 3D models, such as those by Walker et al.'s³³ and Ewart et al.'s³⁴, achieves similar performance on small, curated panels (Walker: 23/27 vs. 23/27; Ewart: 11/14 vs. 13/14). However, their limited scalability constrains their utility to a broader range of DILI mechanisms. They may perform well on narrow, curated panels, but struggle with unknown mechanisms or mechanisms not captured by the low-dimensional endpoint, as shown in the following comparison. To explore the unique capabilities of 2D transcriptomics vs 3D cytotoxicity assays, we conducted direct compound-level comparisons with larger 3D screening studies: Vorrink et al.²⁶ and Fäs et al.³⁵. In Vorrink et al., 3D cytotoxicity uniquely detected 3 compounds (Fialuridine, Methotrexate, Trazodone), all linked to cytotoxic effects that result in acute cell death. Conversely, our model uniquely flagged 10 compounds—including fluconazole, phenytoin, and zileuton—associated with immune activation, metabolic stress, or enzyme modulation, which are not readily captured by viability endpoints. A similar pattern emerged in the Fäs et al. study: 3D cytotoxicity exclusively identified 4 compounds (e.g., Haloperidol, Fialuridine) whose toxicities depend on structural or metabolic context. Our model uniquely identified 5 compounds (e.g., Cimetidine, Fluconazole, Ximelagatran) marked by subtle transcriptional responses rather than overt cell death. These comparisons highlight a key limitation of endpoint-defined models: while effective in narrow contexts, they struggle with broader chemical and mechanistic diversity. Our transcriptomic approach, by contrast, offers systems-level resolution that generalizes across DILI pathways—not only detecting known cytotoxic responses but also uncovering less immediate, non-

lethal mechanisms often missed by traditional assays. While 3D systems offer physiological depth, their reliance on fixed endpoints and lower throughput constrains their utility for a broader range of DILI mechanisms (**Suppl. File S4**).”

Figure 6



2. The presence of only one cell type in the assay (PHH) limits the translational impact as DILI is often a result of complex interplay between many cell types

Response:

We agree with the reviewer that DILI arises from complex interactions involving multiple cell types, including immune and stromal components. While our current

assay uses PHHs—widely regarded as the gold standard—they are nonetheless a monoculture system. As such, they cannot capture multicellular processes such as immune-mediated injury. We have now expanded the Discussion section to explicitly acknowledge this limitation and outline a clear path forward: integrating transcriptomic profiling with advanced 3D co-culture systems that include non-parenchymal cells like Kupffer cells and hepatic stellate cells. We believe our current work provides a strong foundation for such future efforts, demonstrating that transcriptomics alone can already capture a broad range of mechanisms, even in monoculture. The revised Discussion now reflects this perspective.

Addition to Manuscript:

Discussion Section:

“Looking ahead, toxicogenomics holds tremendous promise for advancing our understanding and prediction of DILI. Integrating RNA-seq with advanced 3D culture systems—such as spheroids and liver-on-chip platforms—may enable longer drug exposures and the inclusion of non-parenchymal cells like Kupffer cells and hepatic stellate cells. These co-culture models are essential for capturing immune activation, inflammation, and fibrosis—hallmarks of idiosyncratic and chronic DILI that hepatocyte monocultures cannot recapitulate. As long 3D models rely on high-content imaging or cytotoxicity endpoints that capture specific phenotypic responses they remain limited in mechanistic scope and scalability. Their performance is often demonstrated on small, curated compound sets with known mechanisms, which may not translate to broader chemical space. In contrast, transcriptomics provides a scalable, unbiased readout capable of detecting diverse DILI mechanisms, including those not captured by existing assays. As RNA-seq becomes more feasible in physiologically relevant 3D systems, we anticipate a powerful synergy—combining the mechanistic breadth of transcriptomics with the physiological relevance of 3D models. Our work lays the foundation for such integration, demonstrating that transcriptomics alone can robustly capture DILI risk across a wide range of mechanisms.”

3. The detail regarding the models used for analysis is limited (presumably for commercial reasons) which limits the value of the paper to the reader

Response:

We thank the reviewer for this important comment. We have made every effort to ensure all the modeling process is fully transparent and reproducible. The manuscript now includes a detailed description of the ToxPredictor model, including the derivation of pathway-level signatures, training strategy, cross-validation setup, model selection process, and dose-based safety margin calculation. We benchmarked 193 model configurations across 8 algorithm classes and selected Random Forest based on performance and stability.

In addition, all core model components—including code, processed training data, all validation data, and reproducibility notebooks—are openly available via GitHub, and a dedicated S3 data bucket. These resources allow reproducibility of training, cross-validation, and prediction pipeline end-to-end. We hope this level of transparency

ensures the paper remains valuable to the community. These updates are now detailed in the revised Results, Methods, and Data & Code Availability sections.

Addition to Manuscript:

Results section

“*ToxPredictor*, a machine learning model trained on the *DILImap* foundation, predicts DILI risk from pathway-level transcriptional signatures. These signatures were derived through enrichment analysis (WikiPathways, adj. $p < 0.05$) of genes differentially expressed between compound- and DMSO-treated samples using DESeq2, computed for each dose of every compound in *DILImap* (**Figure 1B**; see **Methods**)

For model training, we used only compounds with unambiguous DILI labels, resulting in a high-confidence training set of 111 DILI+ (*Withdrawn, Known, Likely*) and 52 DILI- (*No DILI*), while the remaining training data were held out to assess model robustness. To ensure high-confidence DILI labels, we further restricted training to drug concentrations tested at more than 20x of their clinical C_{max} to reduce the risk of false-negative labeling for DILI+ compounds that may appear safe at lower doses. For 5-fold cross-validation, we applied stratified, compound-level splitting to ensure that all doses and replicates of a given compound were held out together in each fold, mimicking real-world generalization to unseen compounds. From 193 tested configurations across eight model classes, a *Random Forest* classifier was selected based on its strong performance, achieving high AUC and inter-fold consistency (**Suppl. Fig. S4**).

The final model is an ensemble of 30 Random Forests trained across cross-validation folds to enhance generalization and prediction stability. The ensemble size was chosen based on empirical benchmarking that showed stable test AUC and consistency between models (**Suppl. Fig. S5**). By estimating DILI probabilities across dose levels, *ToxPredictor* enables calculation of drug *safety margins*, defined as the ratio between the first predicted DILI dose (i.e., the lowest dose with predicted probability > 0.7) and the maximum plasma concentration (C_{max}) at therapeutic levels. This provides a transcriptomics-based surrogate of the clinical therapeutic window. A safety margin threshold of 80 provides an actionable classification into high- and low-risk compounds. The probability threshold of 0.7 and MOS cutoff of 80 were both optimized on held-out training data to reach performance plateaus while minimizing false positives. Our selected MOS threshold of 80, while on the higher end of literature-reported ranges (10–100)^{17,23,28,33,35}, reflects the greater sensitivity of transcriptomic assays compared to cytotoxicity or mechanistic readouts. Since transcriptional changes often occur at lower doses—before overt toxicity—a higher cutoff is needed to avoid false positives and maintain specificity in a transcriptome-based model (**Suppl. Fig. S5**). Among available exposure measures, we used total C_{max} instead of free C_{max} due to its broader availability across compounds. Both measures yielded nearly identical predictive performance and highly correlated MOS values (**Suppl. Fig. S6**).

All model selection, hyperparameter tuning, and threshold optimization were performed exclusively on the training data. For final evaluation, we used a fully

independent blind-validation set of 51 compounds (33 DILI+, 14 DILI-, and 4 unknowns), profiled in a separate experiment using separate plates and sequencing runs. This set was withheld from all stages of model development. Compound selection for the validation study was finalized prior to training and intentionally enriched for withdrawals and recent clinical failures. The four unknowns represent compounds currently in clinical use or trials without confirmed DILI liability (**Suppl. Table S1**).

In blind validation, the model achieved 88% sensitivity, correctly identifying 29 of the 33 DILI+ compounds, and 100% specificity, with all 14 DILI- compounds correctly classified as safe (**Figure 1C**)."

Methods section

See all yellow-highlighted paragraphs in the paper.

Suppl. Figures S3

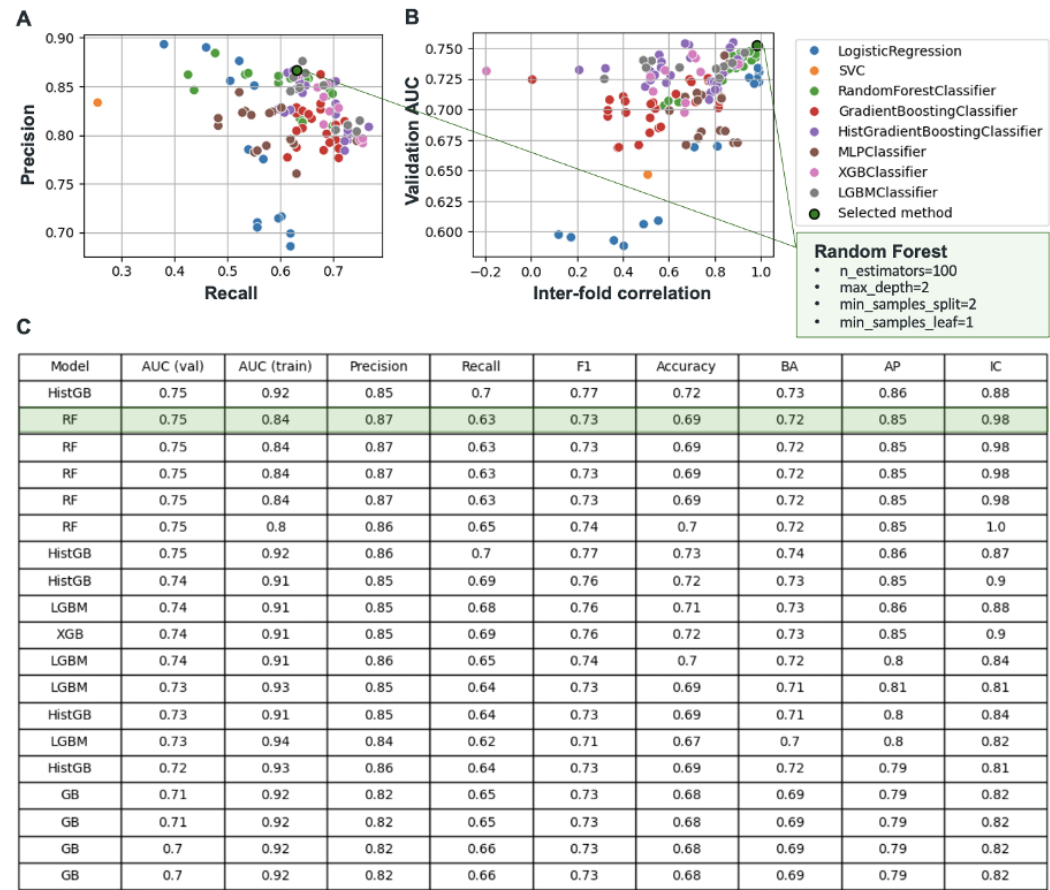


Figure S4. Comparison of model performance across classifiers and hyperparameter configurations. **A.** Precision–recall tradeoff across all 193 tested models, color-coded by classifier type. **B.** Validation AUC vs. inter-fold correlation, illustrating the balance between predictive performance and model stability. **C.** Summary of top-performing models, sorted by validation AUC, showing the following metrics: Area Under the ROC Curve on validation/training folds, Precision, Recall, F1 score, Accuracy, Balanced Accuracy (BA), Average Precision (AP), Inter-fold correlation (correlation of predictions across cross-validation folds on holdout data). The selected model, a Random Forest (RF) with `n_estimators=100`, `max_depth=2`, `min_samples_split=2`, and `min_samples_leaf=1`, is highlighted in green. This configuration was chosen for its strong validation AUC (0.75), low overfitting (AUC gap of 0.09), and high inter-fold correlation (0.98), indicating both robustness and generalizability.

Data and code availability section

Results: All results, reproducibility notebooks, and setup instructions are available at dilimap.org/review-dUFZuIWv8k7bERJ3FQs4. This link will be updated to dilimap.org upon publication. All notebooks can also be downloaded as a ZIP archive [here](#).

Code: The full Python implementation is publicly available at github.com/Cellarity/DILImap

Data: All datasets and trained models are accessible via an S3 bucket, enabling seamless integration with Jupyter notebooks or scripts. Reviewers can use the credentials below for access; the data will be made openly available and deposited in GEO under a Creative Commons license upon publication. A detailed dataset description is available [here](#). Please note, that raw training data (raw counts) are proprietary and available upon request, subject to data-sharing agreements, while processed training data (pathway-level signatures used as model input) are provided to ensure full reproducibility of model training and all figures.

Credentials to access data:

```
AWS_ACCESS_KEY_ID = AKIAZF4LMIKE0ACQPVEU
AWS_SECRET_ACCESS_KEY = CCUSUm/oT5B0A/T7+Lum74Zkq6a85o64Btj+1WRx
```

Reviewer #3 (remarks to the author)

Bergen et al. presented DILImap, a large-scale human toxicogenomics resource, along with ToxPredictor, a Random Forest-based framework for predicting drug-induced liver injury (DILI). The authors present a compelling case for their approach, highlighting the limitations of current DILI prediction methods and emphasizing the potential of DILImap and ToxPredictor to address these challenges. The authors demonstrated the model's ability to achieve high sensitivity and specificity in validation using an independent dataset, outperforming state-of-the-art pre-clinical models. Notably, their approach also flagged previously overlooked failed trials due to DILI, such as Evobrutinib, TAK-875, and BMS-986142. While the study presents an important resource and framework for DILI prediction, several concerns warrant clarification.

1. The numbers of drugs are not consistent and confusing across the manuscript and supplementary data files. While DILImap was claimed to include 300 drugs, the Results section mentions 177 DILI+ and 93 DILI- in the training set and 32 DILI+ and 14 DILI- in the validation data, which are 316 drugs. Supplementary Table shows a total of 299 drugs "DILIMAP_RESULTS".

Response:

Thank you for pointing out the confusion regarding inconsistent numbers. We identified that the *DILIMAP_RESULTS* file was missing *Chlorpromazine*, our positive control, which led to the discrepancy. We have now corrected this and carefully reviewed all instances in the manuscript and supplementary data to ensure consistency. The 177 and 93 reported previously were referring to training data points, not compound numbers.

We confirm that the DILImap dataset contains exactly **300 compounds**, comprising:

- **Training set:** 180 DILI+ and 69 DILI- compounds (total 249)
- **Validation set:** 33 DILI+, 14 DILI-, and 4 without confirmed DILI liability (total 51)

All numbers and references to dataset composition have been updated for clarity and consistency across the manuscript and supplementary materials.

Changes made to the manuscript:

Results section

"The training dataset includes 249 compounds (111 DILI+, 52 DILI-, 17 unlikely DILI, 69 idiosyncratic DILI) for cross-validation. For blind validation, a separate experiment was conducted using an independent set of 51 compounds (33 DILI+, 14 DILI-, and 4 with unknown labels, including real-world clinical failures). This carefully curated dataset provides a robust foundation for predictive modeling and mechanistic insights into DILI."

2. In the modeling process, it is unclear whether the cross-validation mode was done for using all drugs in DILImap or only the training data.

Response:

Thank you for this important clarification request. We confirm that all cross-validation, model selection, hyperparameter tuning, and threshold optimization were performed exclusively on the training set. The validation set was kept fully independent throughout model development and was used only for final evaluation. It was generated in a separate experiment, ensuring no overlap or data leakage between training and validation (as now clarified in the *Results* section).

Additions to manuscript:

Results section

“For model training, we used only compounds with unambiguous DILI labels, resulting in a high-confidence training set of 111 DILI+ (*Withdrawn, Known, Likely*) and 52 DILI– (*No DILI*), while the remaining training data were held out to assess model robustness.

[...]

All model selection, hyperparameter tuning, and threshold optimization were performed exclusively on the training data. For final evaluation, we used a fully independent blind-validation set of 51 compounds (33 DILI+, 14 DILI–, and 4 unknowns), profiled in a separate experiment using separate plates and sequencing runs. This set was withheld from all stages of model development. Compound selection for the validation study was finalized prior to training and intentionally enriched for withdrawals and recent clinical failures. The four unknowns represent compounds currently in clinical use or trials without confirmed DILI liability (**Suppl. Table S1**).”

3. No details on how the training and validation datasets were constructed/split are given. What approach was taken to minimize potential biases in the validation set? This information would be particularly important with the claimed 100% specificity.

Response:

We fully agree that clarity on dataset construction is critical. The training and validation datasets were not created by random splitting, but were derived from two completely independent experiments. The validation set was profiled in a separate experimental batch using different plates, cell passages, and sequencing runs. Compound selection for the validation study was finalized prior to model development and were enriched for withdrawals and recent clinical failures.

In cross-validation, data were using stratified, compound-level splitting—ensuring that all samples from the same compound (across doses and replicates) were held out together in each fold. This simulates a realistic generalization scenario in which

the model must predict DILI risk for entirely unseen compounds. All model selection, hyperparameter tuning, and threshold optimization were restricted to this training data, and the independent validation set was used only for final evaluation.

The reported 100% specificity refers specifically to predictions made on the independent validation set. In cross-validation, which also includes compounds with ambiguous or borderline DILI labels, the model achieved a specificity of 88%.

We have made a few additions to the *Methods* and *Results* sections for clarification.

Additions to manuscript:

Results section

“The training dataset includes 249 compounds (111 DILI+, 52 DILI-, 17 unlikely DILI, 69 idiosyncratic DILI) for cross-validation. For blind validation, a separate experiment was conducted using an independent set of 51 compounds (33 DILI+, 14 DILI-, and 4 with unknown labels, including real-world clinical failures). This carefully curated dataset provides a robust foundation for predictive modeling and mechanistic insights into DILI.

[...]

For 5-fold cross-validation, we applied stratified, compound-level splitting to ensure that all doses and replicates of a given compound were held out together in each fold, mimicking real-world generalization to unseen compounds.

[...]

For final evaluation, we used a fully independent blind-validation set of 51 compounds (33 DILI+, 14 DILI-, and 4 unknowns), profiled in a separate experiment using separate plates and sequencing runs. This set was withheld from all stages of model development. Compound selection for the validation study was finalized prior to training and intentionally enriched for withdrawals and recent clinical failures. The four unknowns represent compounds currently in clinical use or trials without confirmed DILI liability (**Suppl. Table S1**).”

Methods section

“Each model was evaluated using a cross-validation strategy, ensuring that all samples from a single compound were held out together in each fold to prevent data leakage. This compound-level splitting reflects a realistic generalization scenario for novel compound prediction.”

4. Despite extensive comparisons with various assays, the study does not benchmark ToxPredictor against other computational models, such as DILIPredictor, DILIGeNN, etc, for the same drugs.

Response:

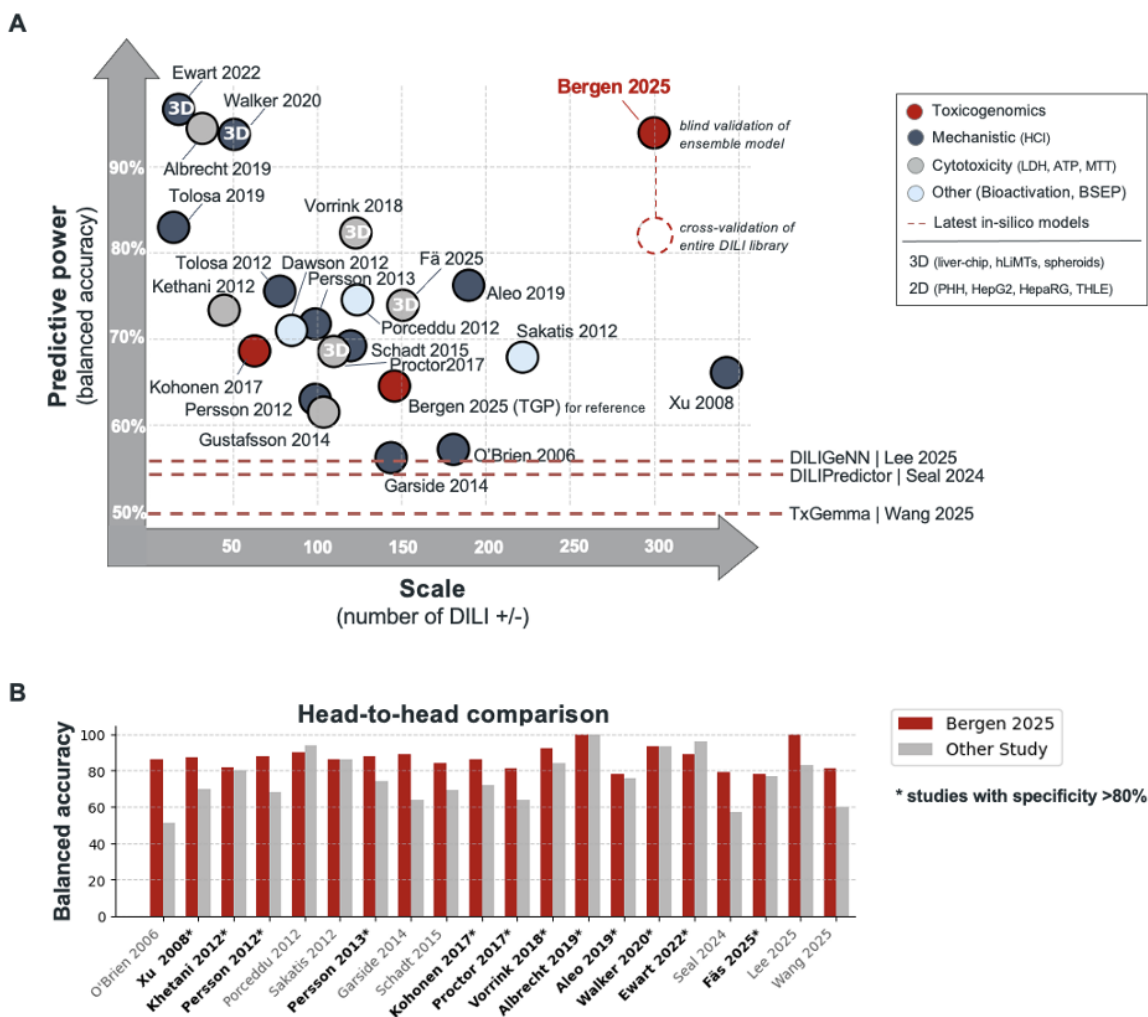
We appreciate the reviewer's suggestion to include a comparison with other in silico models. In the revised manuscript, we now benchmark ToxPredictor against three recent state-of-the-art computational models— *DILIPredictor* , *DILIGeNN* and *TxGemma*—using the same drug set. This comparison is included in both the *Results* section and *Figure 6 (Panels A and B)*. ToxPredictor consistently outperforms these models, particularly in specificity, which is critical for minimizing false safety in DILI prediction.

Additions to manuscript:

Result section

Structure-based in silico models such as TxGemma³¹, DILIGeNN³² and DILIPredictor³³ consistently underperform compared to in vitro–based approaches. To assess real-world generalizability, we evaluated them on 314 independent compounds (45 DILI+, 269 DILI–) primarily annotated via LiverTox (scores A/B as DILI+, E as DILI–); TxGemma was also tested on an expanded set (143 DILI+, 536 DILI–). All showed limited specificity: DILIGeNN (84% sensitivity, 28% specificity), DILIPredictor (80%, 29%), and TxGemma-27B (57%, 37%). These findings are slightly below the balanced accuracy of 0.59 reported by Seal et al. (2024)³³ for DILIPredictor. On a shared benchmark set, TxGemma reached 63% sensitivity (39/62) and 57% specificity (20/35), while our model achieved 76% sensitivity (47/62) and 86% specificity (30/35). DILIGeNN showed perfect sensitivity (5/5) at moderate specificity (2/3), while our model reached 100% on both (5/5 and 3/3). DILIPredictor reached complete sensitivity (23/23) but at the expense of poor specificity (1/7), while ToxPredictor maintained high sensitivity (20/23) at markedly higher specificity (5/7). Low specificity is a key limitation of structure-based models, which lack biological context and tend to over-call toxicity. This results in false positives for commonly prescribed drugs with no risk of hepatotoxicity, such as biotin (DILIPredictor), vitamin D (DILIGeNN), and pemetrexed (all three models). Moreover, they provide only binary outputs, without dose or mechanistic insight. In contrast, transcriptomics enables dose-resolved predictions, mechanistic interpretability, and safety margin estimation—critical for evaluating toxic liabilities and guiding follow-up experiments (**Suppl. Figure S10; Suppl. File S4**).

Figure 6



5. No rationale or supporting data were presented when choosing random forest as the algorithm.

Response:

Thank you for flagging this. We have now clarified the rationale for selecting Random Forest and provided supporting benchmarking data. Across 193 model configurations spanning 8 model classes, Random Forest consistently demonstrated high test AUC and strong inter-fold correlation on held-out compounds—indicating both predictive accuracy and model stability. While gradient

boosting models achieved similar AUCs, Random Forest exhibited lower variance across splits, improved robustness and a reduced tendency to overfit. These findings are now detailed in the revised *Results* and *Methods* sections, with supporting data shown in Suppl. Figure S4.

Additions to manuscript:

Results section

“From 193 tested configurations across eight model classes, a *Random Forest* classifier was selected based on its strong performance, achieving high AUC and inter-fold consistency (**Suppl. Fig. S4**).”

Methods section

“To develop ToxPredictor, we evaluated a range of machine learning models, including Logistic Regression, Support Vector Classifier (SVC), Random Forest, Gradient Boosting Classifier, Hist Gradient Bossting Classifier, XGBoost Classifier, LGBM Classifier and Multi-Layer Perceptron (MLP). Each model was evaluated using a cross-validation strategy, ensuring that all samples from a single compound were held out together in each fold to prevent data leakage. This compound-level splitting reflects a realistic generalization scenario for novel compound prediction. [...] The *Random Forest classifier* emerged as the best-performing model overall. While *XGBoost* and *LightGBM* achieved comparable validation AUCs, Random Forest was selected due to its smaller gap between training and validation AUCs, indicating lower overfitting, and its higher inter-fold correlation, reflecting better stability across data splits. These characteristics along with its ability to handle non-linear relationships, its robustness to noise, and its suitability for datasets of moderate size, make Random Forest the most robust choice for generalization to unseen compounds.”

Suppl. Figure S4

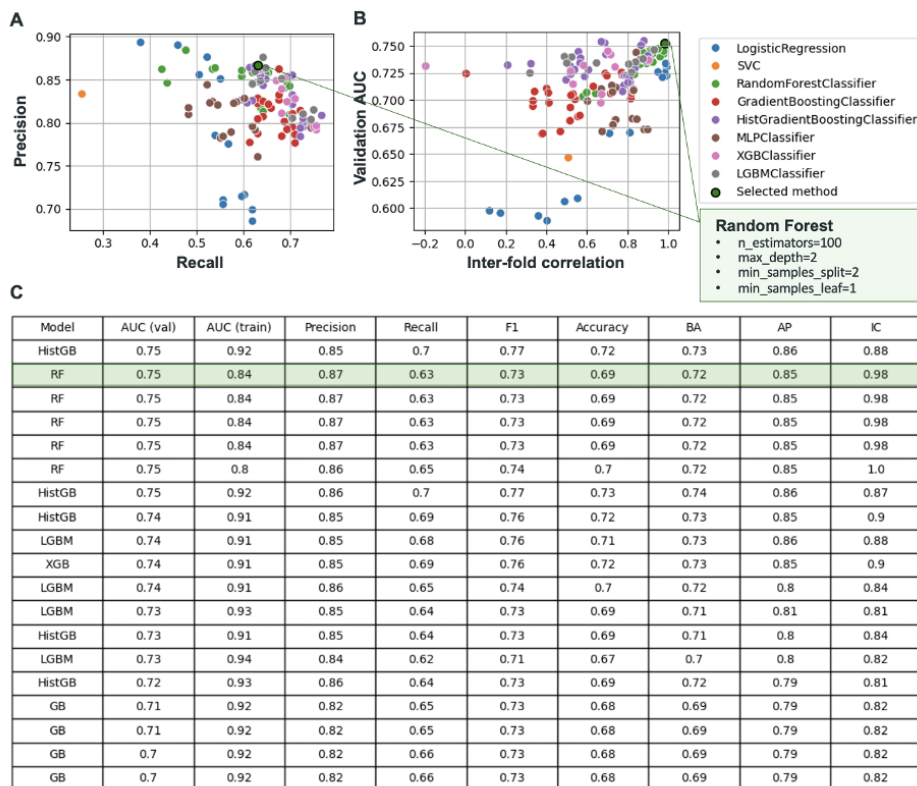


Figure S4. Comparison of model performance across classifiers and hyperparameter configurations. **A.** Precision–recall tradeoff across all 193 tested models, color-coded by classifier type. **B.** Validation AUC vs. inter-fold correlation, illustrating the balance between predictive performance and model stability. **C.** Summary of top-performing models, sorted by validation AUC, showing the following metrics: Area Under the ROC Curve on validation/training folds, Precision, Recall, F1 score, Accuracy, Balanced Accuracy (BA), Average Precision (AP), Inter-fold correlation (correlation of predictions across cross-validation folds on holdout data). The selected model, a Random Forest (RF) with $n_estimators=100$, $max_depth=2$, $min_samples_split=2$, and $min_samples_leaf=1$, is highlighted in green. This configuration was chosen for its strong validation AUC (0.75), low overfitting (AUC gap of 0.09), and high inter-fold correlation (0.98), indicating both robustness and generalizability.

6. Why 30-fold cross-validation was chosen? This fold seems very high considering the small dataset of 300 drugs.

Response:

We appreciate the concern. While 30 folds may seem high given the nominal dataset size of 300 compounds, each drug was profiled across four concentrations, resulting in a larger number of training samples. Our choice was guided by empirical benchmarking: we found that an ensemble of 30 models provided a performance plateau in test AUC and prediction stability. This ensemble approach balances variance reduction and generalization. We have clarified this rationale in the revised *Results* and *Methods* sections, with supporting analyses in *Suppl. Fig. S4*.

Additions to manuscript:

Results section

“The final model is an ensemble of 30 Random Forests trained across cross-validation folds to enhance generalization and prediction stability. The ensemble

size was chosen based on empirical benchmarking that showed stable test AUC and consistency between models (**Suppl. Fig. S5**)."

Methods section

"To ensure predictive robustness, we employed a 30-model Random Forest ensemble using bootstrap aggregation (bagging). Each model was trained on a unique bootstrap sample from the training data, ensuring diversity in training instances and mitigating overfitting. We selected 30 folds to strike a balance between accuracy and ensemble stability. In our benchmarks, ensembles with 30 independent learners consistently achieved smooth, reproducible predictions and stable dose-responses. All fold models used the hyperparameters optimized during cross-validation. At prediction time on unseen compounds, model outputs are averaged with equal weighting, producing a consensus decision that mitigated individual model variance, ensuring robust performance by combining the strengths of each base learner, leading to stable and reliable classification of DILI vs. No-DILI. In blind validation, where predictions were made on compounds not seen during training or model selection, the ensemble demonstrated high AUC, dose consistency, and biological plausibility, confirming its utility for real-world DILI risk assessment."

Suppl. Figure S4 – panel C

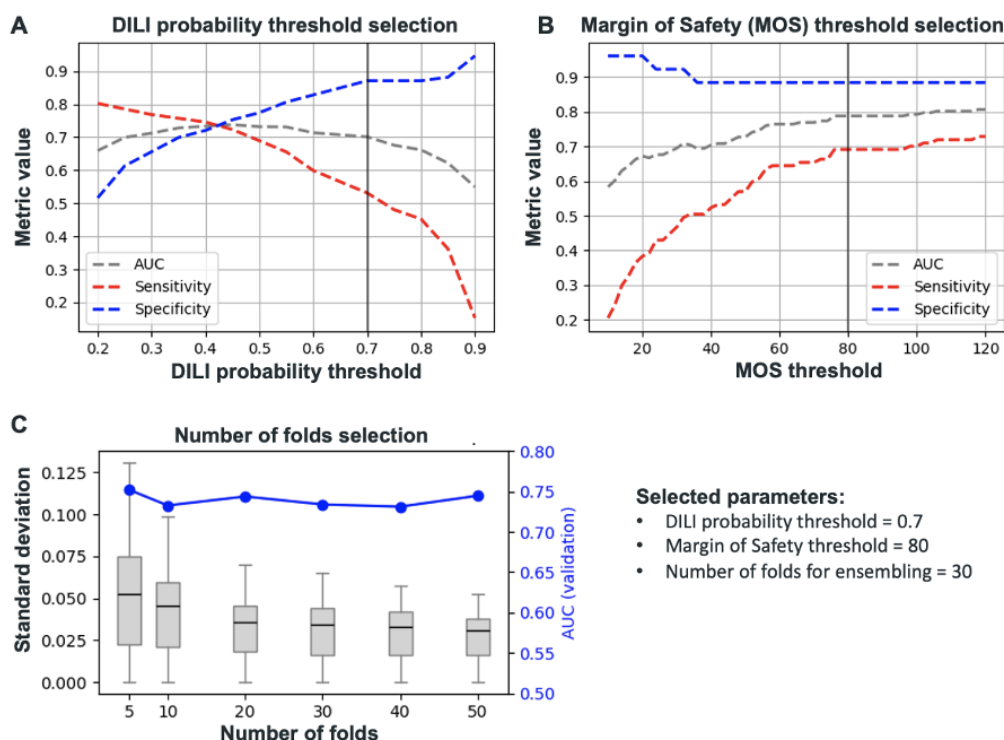


Figure S5. Selection of model threshold parameters based on performance tradeoffs. **A.** DILI probability threshold selection based on sensitivity, specificity, and AUC. A threshold of 0.7 was chosen as it achieves near-plateaued high specificity at acceptable sensitivity. Metric values are computed per-compound/dose basis. Note, that the evaluation metrics here are computed on a per-compound-and-dose basis without inclusion of Cmax. **B.** Margin of Safety (MOS) threshold of 80 was selected for its high plateaued sensitivity and favorable AUC, while preserving high specificity. **C.** Selection of the number of cross-validation folds for model ensembling. Increasing the number of folds reduces the standard deviation across runs, with 30 folds offering a stable tradeoff between ensemble consistency and AUC (validation).

7. An explanation of each column in supplementary tables would be useful.

Response:

Thank you for the suggestion. We have added clear column descriptions for all supplementary tables to improve clarity and interpretability.

8. Codes and data files were available through figshare and code ocean but not completely with many missing data files (e.g. DILlmap_deseq2.h5ad). Please ensure all data files are provided.

Response:

See response below.

9. The authors provided online documentation for installing and using the package; however, the installation instructions did not work with the current version available in their private GitHub repository. While most code and data were accessible, installing the package was challenging. Additionally, some data files were missing, preventing the successful execution of certain scripts.

Response:

Thank you for raising this. As part of our original submission, we used CodeOcean to ensure reproducibility during peer review, with the intention of releasing the full GitHub repository upon acceptance of the manuscript. However, in response to the requests from multiple reviewers—and to facilitate immediate and unrestricted access—we have now made the GitHub repository publicly available. The code is installable via PyPI and includes detailed instructions and reproducibility notebooks. Relevant links and installation details are now included in the *Code and Data Availability* section.

Additions to manuscript:

Data and code availability section

Results: All results, reproducibility notebooks, and setup instructions are available at dilimap.org/review-dUFZulWv8k7bERJ3FQs4. This link will be updated to dilimap.org upon publication. All notebooks can also be downloaded as a ZIP archive [here](#).

Code: The full Python implementation is publicly available at github.com/Cellarity/DILlmap

Data: All datasets and trained models are accessible via an S3 bucket, enabling seamless integration with Jupyter notebooks or scripts. Reviewers can use the credentials below for access; the data will be made openly available and deposited

in GEO under a Creative Commons license upon publication. A detailed dataset description is available [here](#). Please note, that raw training data (raw counts) are proprietary and available upon request, subject to data-sharing agreements, while processed training data (pathway-level signatures used as model input) are provided to ensure full reproducibility of model training and all figures.

Credentials to access data:

```
AWS_ACCESS_KEY_ID = AKIAZF4LMIKE0ACQPVEU  
AWS_SECRET_ACCESS_KEY = CCUSUm/oT5B0A/T7+Lum74Zkq6a85o64Btj+lWRx
```

Point-by-point Response to the Reviewers

Reviewer #1 (remarks to the author)

The authors have successfully addressed the concerns and questions I raised. It is appreciated that the entire codebase, data, and user guide are now accessible via GitHub, making it easy for users to install the PyPI package. Furthermore, the missing method section is now available in the supplementary file.

I also appreciate the efforts of the authors to clarify the application of free Cmax or total Cmax. The author demonstrated that neither free Cmax nor total Cmax significantly impacted their model's ability to predict DILI. This is very interesting! It is also good to see that the authors discussed the combination of emerging 3D liver models and DILImap in the updated manuscript. This is very important to improve the future application of New Approach Methodologies for drug development.

Response:

Thank you very much for your thoughtful and encouraging feedback. We are truly grateful for your careful review and greatly appreciate your recognition of our revisions and contributions.

Reviewer #2 (remarks to the author)

I am satisfied with the authors comments

Response:

Thank you very much for your comments and for the time you invested in reviewing our work. We greatly appreciate your feedback and are glad that our revisions addressed your concerns.

Reviewer #3 (remarks to the author)

The authors have addressed most of my previous comments sufficiently, and I appreciate the clarifications provided in the revision. I have only a few minor follow-up comments.

1. The final rationale for prioritizing Random Forest over other models with similar AUCs could still be made clearer. It would help to briefly explain how the evaluation metrics such as AUC, inter-fold correlation, dose response, precision, and recall were weighted or prioritized in arriving at the final choice.

2. In Figure S4C, only performance metrics are shown without the corresponding **hyperparameters**. Since it is not feasible to include various hyperparameters in this table, I suggest providing another supplementary file with hyperparameter details for these model configurations.

3. Regarding the CV clarification, I understand from the revised text that 5-fold stratified compound-level cross-validation (30-fold in the original submission) was used for model training and evaluation. The compound-level approach, described as “all samples from a single compound were held out together,” is clear and appropriate for preventing data leakage. The exact number of folds better be stated explicitly in the Methods section as well.

4. The current description in the final RF ensemble model, “The final model is an ensemble of 30 Random Forests trained across cross-validation folds to enhance generalization and prediction stability,” is confusing. It would be clearer to use the original wording “30 models” or “30 ensemble members,” rather than the current “30 folds.” (here and in the Methods as well).

Response:

Thank you very much for your constructive and detailed feedback. We greatly appreciate your thoughtful follow-up comments and have addressed them as follows:

1. We expanded the rationale for prioritizing Random Forests, clarifying how we balanced AUC, inter-fold correlation, dose–response consistency, precision, and recall when selecting the final model.
2. We provided the corresponding hyperparameters for the model configurations shown in Figure S4C in Supplementary File S4.
3. We explicitly stated in the Methods that 5-fold stratified compound-level cross-validation was used, ensuring clarity and consistency.
4. We revised the ensemble description to “30 Random Forest models (ensemble members),” as suggested.

We are grateful for your careful review and all your suggestions, which have significantly improved the clarity and rigor of the manuscript.

Additions to Manuscript:

Main

From 193 tested configurations across eight model classes, we selected a Random Forest classifier for its strong validation AUC, minimal overfitting, and highest consistency across folds. These properties, combined with its interpretability, motivated its choice over more complex boosting and deep learning models (**Suppl. Fig. S4**).

Methods

Model selection strategy

- **Generalizability:** prioritize high validation AUC with low train–validation gap to avoid overfitting
→ Best models had AUC ≥ 0.74 (RF, HistGB, LGBM, XGB); RF showed the lowest gap (~ 0.09 vs. $0.16\text{--}0.20$ for GBMs).
- **Stability:** require high inter-fold correlation for consistent predictions across folds
→ RF achieved $0.98\text{--}1.0$, higher than GBMs ($0.8\text{--}0.9$).
- **Clinical utility:** balance precision (reduce false positives) and recall (reduce false negatives)
→ RF performance was comparable to other top models.
- **Interpretability:** prefer simpler, easier-to-interpret models at similar performance
→ RF is more interpretable than boosting methods.

Final choice: Random Forest was selected as the most balanced and robust model, combining strong validation AUC, minimal overfitting, high stability, and interpretability. Although XGBoost and LightGBM reached similar AUCs, they showed greater overfitting (train-validation AUC gap) and lower stability (inter-fold correlation). These characteristics along with its ability to handle non-linear relationships, its robustness to noise, and its suitability for datasets of moderate size, make Random Forest the most robust choice for generalization to unseen compounds.

Main

The final model is an ensemble of 30 Random Forest models (ensemble members) trained on different cross-validation splits, which together enhance generalization and prediction stability

Methods

Each model was evaluated using a 5-fold stratified compound-level cross-validation strategy.